Post-Selection Sensitivity in a GHZ-Type Mermin Statistic: An In-Silico Audit

GHZ 型 Mermin 统计量的后选择敏感性：仿真审计

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Scope / 范围: `medium-v10` in-silico GHZ-type audit / `medium-v10` 仿真 GHZ 型审计（非硬件复刻）
Source code & data / 源代码与数据： https://github.com/tomnattle/chain-explosion-model
Clone this repository and run §6 from the **repository root**. / 请克隆该仓库，在**仓库根目录**执行第 6 节命令。

ENGLISH

The Greenberger-Horne-Zeilinger (GHZ) state is historically regarded as the "smoking gun" of quantum non-locality. **Computational scope:** this HTML note summarizes a `medium-v10` **simulator audit** of a GHZ-type Mermin statistic (local phase-locked sources), not a hardware rerun. Within that model, large F tracks low retention under gating; random matched-retention controls fail to reproduce the gated lift.

中文对照

Greenberger-Horne-Zeilinger (GHZ) 态在历史上被视为量子非局域性的“冒烟枪”证据。**计算范围：** 本文为仓库 `medium-v10` **模拟器**中的 GHZ 型 Mermin 统计量审计摘要，并非某一硬件实验的逐台复刻。该模型下，高 F 与门控导致的低保留率绑定；同保留率随机对照无法复现门控抬升。

1. Introduction: Audit Scope and Motivation / 引言：审计范围与动机

In-silico finding (this note): within `medium-v10`, we summarize sensitivity of a GHZ-type statistic to gating and

仿真内结论（本文）：在 `medium-v10` 内概括 GHZ 型统计量对门控与保留的敏感性（§2–§3）。**解释性假说（未证明）：**实验

retention (§ 2– § 3). **Interpretive hypothesis (not proved here):** lab Bell/GHZ *narratives* often under-report how much correlation moves when inclusion rules change. The next blocks are **research suspicion**—motivated by audit culture, **not** established by the V10.4 curve alone.

室贝尔/GHZ *叙事*常低估纳入规则变动时关联的位移。下文为**研究怀疑**——由审计文化启发，**非**仅凭 V10.4 曲线可证。

Hypothesis: violations are often *read* as clean non-locality while discard sensitivity stays off the ledger. **Hypothesis:** in a continuous-field picture, “photons” may be better read as **threshold-triggered readout events** than as context-free particles—**a modeling suspicion for debate, not a simulator output.**

假说：违背常被*读作*非局域证书，丢弃敏感性却未入账。**假说：**连续场图景下，“光子”或许更接近**阈值触发的读出事件**——**供辩论的建模怀疑，非模拟器结果。**

Hypothesis: headline “success” in some experiments is **compatible with** heavy post-selection; *if* engineering must budget the whole stream, roadmaps that only advertise peak metrics after aggressive filtering may face a **scaling–accounting tension**. **Conditional suspicion, not a universal lab fact.**

假说：部分实验头条成功**可与**重度后选择相容；*若*工程须为全流买单，仅宣传激进过滤后峰值的路线图或面临**规模化**—**账目**张力。**条件性怀疑，非普适事实。**

2. Methodology: Post-Selection and Selection Tax / 方法论：后选择与选择税

We model the GHZ setup using a medium-wave propagation model (`medium-v10`) with three local phase-locked sources. Instead of forcing a "perfect success," we sweep the gate strength of a soft detector.

- **Selection rule:** events are recorded only if the interference amplitude exceeds a threshold.

我们使用中等波动传播模型（`medium-v10`）对 GHZ 装置进行建模，包含三个局域相位锁定源。我们不再追求“强行成功”，而是扫描软探测器的门控强度。

- **筛选规则：**仅当干涉振幅超过阈值时才记录事件。
- **随机对照：**对每个门控结果，抽取**同保留率**的随机子样本，以区分机制抬升与噪声。

- **Random control:** for each gated outcome, draw a random subsample with the *same retention ratio* to separate mechanism-driven lift from noise.

3. Results: Selection-Retention Trade-off / 结果：筛选-保留率权衡

The V10.4 pipeline shows a direct trade-off between Mermin-style **F** and **data retention** under amplitude gating. High **F** tracks discarding most draws; matched-retention random baselines do not reproduce the gated curve.

V10.4 流程显示：在振幅门控下，Mermin 型 **F** 与**数据保留率**存在直接权衡。高 **F** 与丢弃大多数样本相伴；同保留率的随机基线无法复现门控曲线。

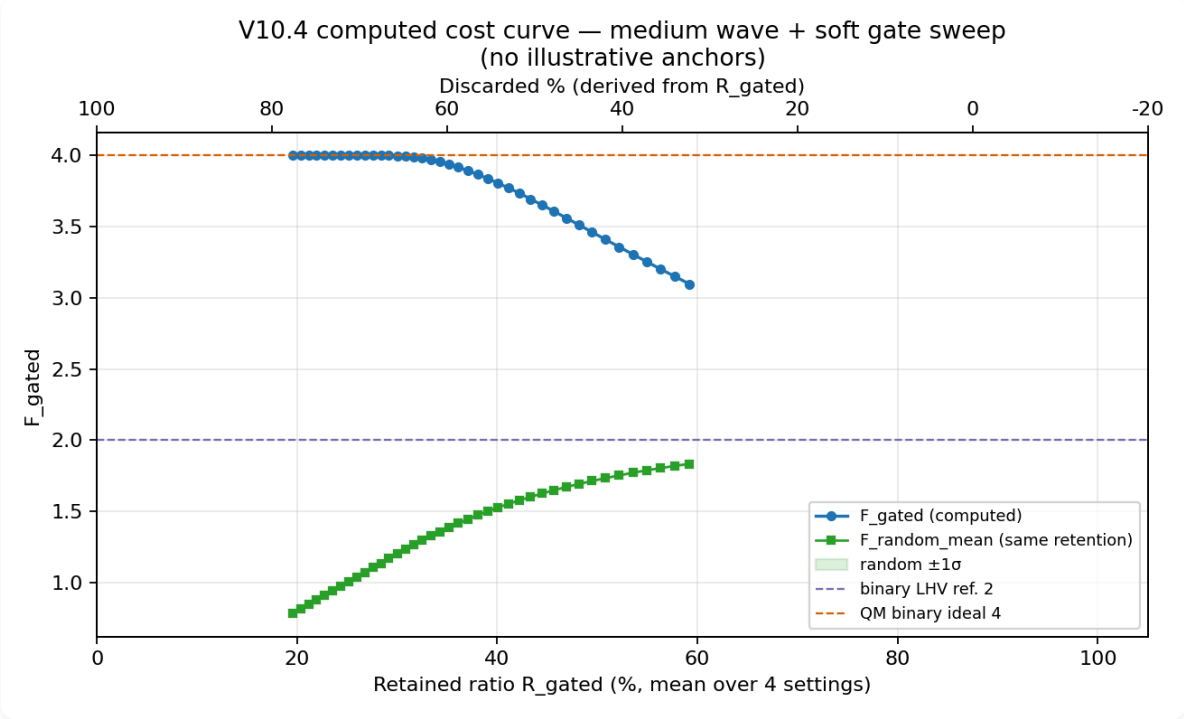

Figure 1: GHZ real cost curve (V10.4). Blue: gated F; green: matched-retention random baseline; gap encodes selection rules. / 图 1：GHZ 真实代价曲线（V10.4）。蓝：门控 F；绿：同保留率随机基线；间距编码筛选规则。

4. Discussion: The "30cm Truth" and the Geometric Illusion / 讨论："30厘米的真相"与几何幻觉

Bell-style work uses the "30 cm truth" as a **scale argument** (~**1 ns** ↔ ~**30 cm** in

贝尔侧工作把"30 厘米的真相"当作**尺度论证**（真空中约 **1 纳秒** ↔ **约 30 厘米**），说

vacuum) for why tiny timing rules move coincidence counts. **This GHZ leg** is *not* a timing-coincidence experiment: it couples correlation to **amplitude gating / retention** in `medium-v10`. The shared structural risk: **analysis rules move mass between kept vs discarded samples**, moving reported metrics.

明微小时间规则如何搬动符合计数。**GHZ 本条*不是*时间符合实验：它在** `medium-v10` **内把关联与振幅门控/保留率耦合。共同结构风险：分析规则在保留与丢弃样本之间搬动质量，从而搬动报告指标。**

4.1 Implications for Quantum Computing (interpretive hypothesis only) / 对量子计算的影响（仅为解释性假说）

No new data. Speculative bridge from in-silico selection to scaling narratives—not a vendor prediction; **not** a claim that large QC has "failed" for this reason.

无新数据。从仿真筛选到规模化叙事的推测桥接——非厂商预言；**不**主张大规模量子计算已因此"失败"。

Hypothesis: *If* demos lean on metrics that spike only after aggressive filtering, a "hollow foundation" **might** be one **among many** ways to narrate scaling friction—**not** an explanation this note verifies. (Contrast: whole-dataset ML vs peak metrics under undisclosed discards.)

假说：*若*演示依赖激进过滤后才冲高的指标，"虚浮地基"**或许**是描述规模化摩擦的**多种叙事之一**——**非**本文已验证的解释。（对照：全量 ML vs 丢弃未披露时的峰值。）

Hypothesis: *If* entanglement-as-reported is often **entangled with** gating geometry, processors might **sometimes** solve **filtering/bookkeeping** alongside Hamiltonian physics—**suspicion about KPI design**, not a denial of every QM prediction.

假说：*若*报告中的纠缠常与门控几何**纠缠**，处理器**有时**或在哈密顿物理之外并行处理**过滤/账目**——**对 KPI 的怀疑**，**非**否定全部量子预测。

> **Interpretive hypothesis (rhetorical).** "If the foundation is hollow, the skyscraper of quantum computing is solving geometry, not entanglement." / **解释性假说（修辞）。**"如果地基是虚幻的，量子计算的摩天大楼解决的是几何问题，而非纠缠问题。"

5. Conclusion: A Boundary-Aware Audit Summary / 结论：边界清晰的审计总结

We do not claim quantum mechanics is "wrong" in its formal predictions, but we treat many foundational *reported* metrics as "unbalanced accounts" unless retention, gates, and denominators are disclosed. We document pairing windows, amplitude gating, and binarization as bookkeeping levers.

我们并不宣称量子力学的形式预测“错误”，但若未披露保留率、门控与分母，许多**报告**指标应被视为“不平账目”。我们记录配对窗口、振幅门控与二值化等平账杠杆。

We invite practitioners to archive the **discarded majority** (or equivalent inclusion logs) for independent audit—not only headline curves.

我们邀请实践者归档**被丢弃的大多数**（或等价的纳入日志）以供独立审计——而非仅 headline 曲线。

6. Reproducibility / 最小复现

Repository / 仓库： https://github.com/tomnattle/chain-explosion-model

EN Run from repo root: `python scripts/explore/ghz_medium_v10/v10_4_real_cost_curve.py` (outputs per script header). Figure in this package: `figures/V10_4_REAL_COST_CURVE.png`.

中 在仓库根目录运行：`python scripts/explore/ghz_medium_v10/v10_4_real_cost_curve.py`（输出见脚本头部）。本包图：`figures/V10_4_REAL_COST_CURVE.png`。

Scope / 范围: `medium-v10` **in-silico** audit — not a hardware GHZ rerun / **仿真**审计，非硬件 GHZ 复刻
Data Availability / 数据可用性: GitHub repo above; Zenodo bundles may mirror this archive / 见上列 GitHub；Zenodo 压缩包可为本文档镜像
Verification / 验证: `scripts/explore/ghz_medium_v10/v10_4_real_cost_curve.py`
Zenodo / DOI: 10.5281/zenodo.19785022 · record
GitHub / 仓库: https://github.com/tomnattle/chain-explosion-model